

The original results for the shortest paths for a forward-only car and for a car with a reverse gear were given by Dubins [41] and Reeds and Shepp [146], respectively. These results were extended and applied to motion planning problems in [16, 175] and independently derived using principles of differential geometry in [183]. The minimum-time motions for a diff-drive were derived by Balkcom and Mason in [5]. The motion planner for a car-like mobile robot based on replacing segments of an arbitrary path with shortest feasible paths for a car, described in Section 13.3.3.2, was described in [82].

The nonlinear control law (13.31) for tracking a reference trajectory for a nonholonomic mobile robot is taken from [121, 157].

13.8 Exercises

Exercise 13.1 In the omnidirectional mobile robot kinematic modeling of Section 13.2.1, we derived the relationship between wheel velocities and chassis velocity in what seemed to be an unusual way. First, we specified the chassis velocity, then we calculated how the wheels must be driving (and sliding). At first glance, this approach does not seem to make sense causally; we should specify the velocities of the wheels, then calculate the chassis velocity. Explain mathematically why this modeling approach makes sense, and under what condition the method cannot be used.

Exercise 13.2 According to the kinematic modeling of Section 13.2.1, each wheel of an omnidirectional robot adds two more velocity constraints on the chassis twist $\mathcal{V}_b$. This might seem counterintuitive, since more wheels means more motors and we might think that having more motors should result in more motion capability, not more constraints. Explain clearly why having extra wheels implies extra velocity constraints, in our kinematic modeling, and which assumptions in the kinematic modeling may be unrealistic.

Exercise 13.3 For the three-omniwheel robot of Figure 13.5, is it possible to drive the wheels so that they skid? (In other words, so that the wheels slip in the driving direction.) If so, give an example set of wheel velocities.

Exercise 13.4 For the four-mecanum-wheel robot of Figure 13.5, is it possible to drive the wheels so that they skid? (In other words, the wheels slip in the

driving direction.) If so, give an example set of wheel velocities.

Exercise 13.5 Replace the wheels of the four-mecanum-wheel robot of Figure 13.5 by wheels with $\gamma = \pm 60^\circ$. Derive the matrix $H(0)$ in the relationship $u = H(0)\mathcal{V}_b$. Is it rank 3? If necessary, you can assume values for ℓ and w .

Exercise 13.6 Consider the three-omniwheel robot of Figure 13.5. If we replace the omniwheels by mecanum wheels with $\gamma = 45^\circ$, is it still a properly constructed omnidirectional mobile robot? In other words, in the relationship $u = H(0)\mathcal{V}_b$, is $H(0)$ rank 3?

Exercise 13.7 Consider a mobile robot with three mecanum wheels for which $\gamma = \pm 45^\circ$ at the points of an equilateral triangle. The chassis frame $\{b\}$ is at the center of the triangle. The driving directions of all three wheels are the same (e.g., along the body $\hat{x}_b$ -axis) and the free sliding directions are $\gamma = 45^\circ$ for two wheels and $\gamma = -45^\circ$ for the other wheel. Is this a properly constructed omnidirectional mobile robot? In other words, in the relationship $u = H(0)\mathcal{V}_b$, is $H(0)$ rank 3?

Exercise 13.8 Using your favorite graphics software (e.g., MATLAB), plot the two planes bounding the set of feasible body twists $\mathcal{V}_b$ for wheel 2 of the three-omniwheel robot of Figure 13.5.

Exercise 13.9 Using your favorite graphics software (e.g., MATLAB), plot the two planes bounding the set of feasible body twists $\mathcal{V}_b$ for wheel 1 of the four-mecanum-wheel robot of Figure 13.5.

Exercise 13.10 Consider a four-omniwheel mobile robot with wheels at the points of a square. The chassis frame $\{b\}$ is at the center of the square, and the driving direction of each wheel is in the direction 90° counterclockwise from the vector from the origin of $\{b\}$ to the wheel. You may assume that the sides of the squares have length 2. Find the matrix $H(0)$. Is it rank 3?

Exercise 13.11 Implement a collision-free grid-based planner for an omnidirectional robot. You may assume that the robot has a circular chassis, so for collision-detection purposes, you need to consider only the (x, y) location of the robot. The obstacles are circles with random center locations and random radii. You can use Dijkstra's algorithm or A^* to find a shortest path that avoids

obstacles.

Exercise 13.12 Implement an RRT planner for an omnidirectional robot. As above, you may assume a circular chassis and circular obstacles.

Exercise 13.13 Implement a feedforward plus proportional feedback controller to track a desired trajectory for an omnidirectional mobile robot. Test it on the desired trajectory $(\phi_d(t), x_d(t), y_d(t)) = (t, 0, t)$ for $t \in [0, \pi]$. The initial configuration of the robot is $q(0) = (-\pi/4, 0.5, -0.5)$. Plot the configuration error as a function of time. You can also show an animation of the robot converging to the trajectory.

Exercise 13.14 Write down the Pfaffian constraints $A(q)\dot{q} = 0$ corresponding to the unicycle model in Equation (13.12).

Exercise 13.15 Write down the Pfaffian constraints corresponding to the diff-drive model in Equation (13.14).

Exercise 13.16 Write down the Pfaffian constraints corresponding to the car-like model in Equation (13.16).

Exercise 13.17 Give examples of two systems that are STLA but not STLC. The systems should not be wheeled mobile robots.

Exercise 13.18 Continue the Taylor expansion in Equation (13.26) to find the net motion (to order ϵ^2) obtained by following g_i for time ϵ , then g_j for time ϵ , then $-g_i$ for time ϵ , then $-g_j$ for time ϵ . Show that it is equivalent to the expression (13.27).

Exercise 13.19 Write down the canonical nonholonomic mobile robot model (13.18) in the chassis-fixed form

$$\mathcal{V}_b = B \begin{bmatrix} v \\ \omega \end{bmatrix},$$

where B is a 3×2 matrix whose columns correspond to the chassis twist associated with the controls v and ω .

Exercise 13.20 Throughout this book, we have been using the configuration spaces $SE(3)$ and $SO(3)$, and their planar subsets $SE(2)$ and $SO(2)$. These are

known as matrix Lie groups. Body and spatial twists are represented in matrix form as elements of $se(3)$ (or $se(2)$ in the plane) and body and spatial angular velocities are represented in matrix form as elements of $so(3)$ (or $so(2)$ in the plane). The spaces $se(3)$ and $so(3)$ correspond to all possible $\dot{T}$ and $\dot{R}$ when T or R is the identity matrix. Since these spaces correspond to all possible velocities, each is called the Lie algebra of its respective matrix Lie group. Let us call G a matrix Lie group and $\mathfrak{g}$ its Lie algebra, and let X be an element of G , and A and B be elements of $\mathfrak{g}$. In other words, A and B can be thought of as possible values of $\dot{X}$ when $X = I$.

Any “velocity” A in $\mathfrak{g}$ can be “translated” to a velocity $\dot{X}$ at any $X \in G$ by pre-multiplying or post-multiplying by X , i.e., $\dot{X} = XA$ or $\dot{X} = AX$. If we choose $\dot{X} = XA$, i.e., $A = X^{-1}\dot{X}$ then we can think of A as a “body velocity” (e.g., the matrix form of a body twist if $G = SE(3)$) and if we choose $\dot{X} = AX$, i.e., $A = \dot{X}X^{-1}$, we can think of A as a “spatial velocity” (e.g., the matrix form of a spatial twist if $G = SE(3)$). In this way, A can be extended to an entire vector field over G . If the extension is obtained by multiplying by X on the left then the vector field is called left-invariant (constant in the body frame), and if the extension is by multiplying by X on the right then the vector field is called right-invariant (constant in the space frame). Velocities that are constant in the body frame, like the vector fields for the canonical nonholonomic mobile robot, correspond to left-invariant vector fields.

Just as we can define a Lie bracket of two vector fields, as in Equation (13.28), we can define the Lie bracket of $A, B \in \mathfrak{g}$ as

$$[A, B] = AB - BA, \quad (13.38)$$

as described in Section 8.2.2 for $\mathfrak{g} = se(3)$. Confirm that this formula describes the same Lie bracket vector field as Equation (13.28) for the canonical nonholonomic vector fields $g_1(q) = (0, \cos \phi, \sin \phi)$ and $g_2(q) = (1, 0, 0)$. To do this, first express the two vector fields as $A_1, A_2 \in se(2)$, considered to be the generators of the left-invariant vector fields $g_1(q)$ and $g_2(q)$ (since the vector fields correspond to constant velocities in the chassis frame). Then take the Lie bracket $A_3 = [A_1, A_2]$ and extend A_3 to a left-invariant vector field defined at all $X \in SE(2)$. Show that this is the same result obtained using the formula (13.28).

The Lie bracket formula in Equation (13.28) is general for any vector fields expressed as a function of coordinates q , while the formula (13.38) is particularly for left- and right-invariant vector fields defined by elements of the Lie algebra of a matrix Lie group.

Exercise 13.21 Using your favorite symbolic math software (e.g., Mathe-

matica), write or experiment with software that symbolically calculates the Lie bracket of two vector fields. Show that it correctly calculates Lie brackets for vector fields of any dimension.

Exercise 13.22 For the full five-dimensional diff-drive model described by Equation (13.14), calculate Lie products that generate two motion directions not present in the original two vector fields. Write down the holonomic constraint corresponding to the direction in which it is not possible to generate motions.

Exercise 13.23 Implement a collision-free grid-based motion planner for a car-like robot among obstacles using techniques from Section 10.4.2. Decide how to specify the obstacles.

Exercise 13.24 Implement a collision-free RRT-based motion planner for a car-like robot among obstacles using techniques from Section 10.5. Decide how to specify obstacles.

Exercise 13.25 Implement the reference point trajectory-tracking control law (13.29) for a diff-drive robot. Show in a simulation that it succeeds in tracking a desired trajectory for the point.

Exercise 13.26 Implement the nonlinear feedforward plus feedback control law (13.31). Demonstrate its performance in tracking a reference trajectory with different sets of control gains, including one set that yields “good” performance.

Exercise 13.27 Write a program that accepts a time history of wheel encoder values for the two rear wheels of a car and estimates the chassis configuration as a function of time using odometry. Prove that it yields correct results for a chassis motion that involves rotations and translations.

Exercise 13.28 Write a program that accepts a time history of wheel encoder values for the three wheels of a three-omniwheel robot and estimates the chassis configuration as a function of time using odometry. Prove that it yields correct results for a chassis motion that involves rotations and translations.

Exercise 13.29 Write a program that accepts a time history of wheel encoder values for the four wheels of a four-mecanum-wheel robot and estimates the chassis configuration as a function of time using odometry. Prove that it yields

correct results for a chassis motion that involves rotations and translations.

Exercise 13.30 Consider the mobile manipulator in Figure 13.24. Write down the Jacobian $J_e(\theta)$ as a 3×3 matrix function of d , x_r , L_1 , and θ_1 . Is the Jacobian rank 3 for all choices of d , x_r , L_1 , and θ_1 ? If not, under what conditions is it not full rank?

Exercise 13.31 Write a simulation for a mobile manipulation controller similar to that demonstrated in Figure 13.24. For this simulation you need to include a simulation of odometry to keep track of the mobile base's configuration. Demonstrate your controller on the same example trajectory and initial conditions shown in Figure 13.24 for good and bad choices of control gains.

Exercise 13.32 Wheel-based odometry can be supplemented by odometry based on an inertial measurement unit (IMU). A typical IMU includes a three-axis gyro, for sensing angular velocities of the chassis, and a three-axis accelerometer, for sensing linear accelerations of the chassis. From a known initial state of the mobile robot (e.g., at rest at a known position), the sensor data from the IMU can be integrated over time to yield a position estimate of the robot. Since the data are numerically integrated once in the case of angular velocities, and twice in the case of linear accelerations, the estimate will drift from the actual value over time, just as the wheel-based odometry estimate will.

In a paragraph describe operating conditions for the mobile robot, including properties of the wheels' interactions with the ground, where IMU-based odometry might be expected to yield better configuration estimates, and conditions where wheel-based odometry might be expected to yield better estimates. In another paragraph describe how the two methods can be used simultaneously, to improve the performance beyond that of either method alone. You should feel free to do an internet search and comment on specific data-fusion tools or filtering techniques that might be useful.

Exercise 13.33 The KUKA youBot (Figure 13.25) is a mobile manipulator consisting of a 5R arm mounted on an omnidirectional mobile base with four mecanum wheels. The chassis frame $\{b\}$ is centered between the four wheels at a height $z = 0.0963$ m above the floor, and the configuration of the chassis

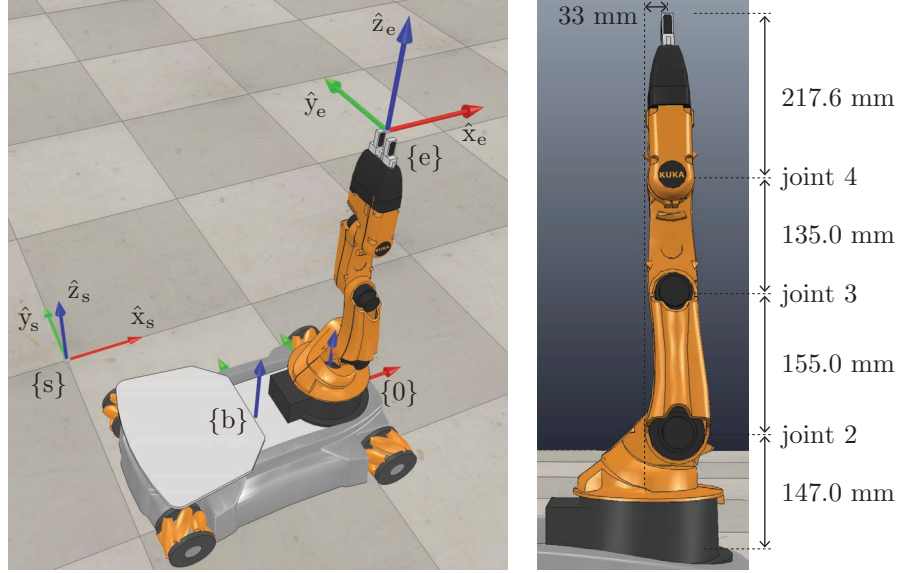


Figure 13.25: (Left) The KUKA youBot mobile manipulator and the fixed space frame $\{s\}$, the chassis frame $\{b\}$, the arm base frame $\{0\}$, and the end-effector frame $\{e\}$. The arm is at its zero configuration. (Right) A close-up of the arm at its zero configuration. Joint axes 1 and 5 (not shown) point upward and joint axes 2, 3, and 4 are out of the page.

relative to a fixed space frame $\{s\}$ is

$$T_{sb}(q) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 & x \\ \sin \phi & \cos \phi & 0 & y \\ 0 & 0 & 1 & 0.0963 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where $q = (\phi, x, y)$. The kinematics of the four-wheeled mobile base is described in Figure 13.5 and the surrounding text, where the front-back distance between the wheels is $2\ell = 0.47$ m, the side-to-side distance between the wheels is $2w = 0.3$ m, and the radius of each wheel is $r = 0.0475$ m.

The fixed offset from the chassis frame $\{b\}$ to the base frame of the arm $\{0\}$ is

$$T_{b0} = \begin{bmatrix} 1 & 0 & 0 & 0.1662 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.0026 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

i.e., the arm base frame $\{0\}$ is aligned with the chassis frame $\{b\}$ and is displaced by 166.2 mm in $\hat{x}_b$ and 2.6 mm in $\hat{z}_b$. The end-effector frame $\{e\}$ at the zero configuration of the arm (as shown in Figure 13.25) relative to the base frame $\{0\}$ is

$$M_{0e} = \begin{bmatrix} 1 & 0 & 0 & 0.0330 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0.6546 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

You can ignore the arm joint limits in this exercise.

- (a) Examining the right-hand side of Figure 13.25 – and keeping in mind that (i) joint axes 1 and 5 point up on the page and joint axes 2, 3, and 4 point out of the page, and (ii) positive rotation about an axis is by the right-hand rule – either confirm that the screw axes in the end-effector frame $\mathcal{B}_i$ are as shown in the following table:

i	ω_i	v_i
1	$(0, 0, 1)$	$(0, 0.0330, 0)$
2	$(0, -1, 0)$	$(-0.5076, 0, 0)$
3	$(0, -1, 0)$	$(-0.3526, 0, 0)$
4	$(0, -1, 0)$	$(-0.2176, 0, 0)$
5	$(0, 0, 1)$	$(0, 0, 0)$

or provide the correct $\mathcal{B}_i$.

- (b) The robot arm has only five joints, so it is incapable of generating an arbitrary end-effector twist $\mathcal{V}_e \in \mathbb{R}^6$ when the mobile base is parked. If we are able to move the mobile base and the arm joints simultaneously, are there configurations θ of the arm at which arbitrary twists are not possible? If so, indicate these configurations. Also explain why the configuration $q = (\phi, x, y)$ of the mobile base is irrelevant to this question.
- (c) Use numerical inverse kinematics to find a chassis and arm configuration (q, θ) that places the end-effector at

$$X(q, \theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1.0 \\ 0 & -1 & 0 & 0.4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

You can try $q_0 = (\phi_0, x_0, y_0) = (0, 0, 0)$ and $\theta_0 = (0, 0, -\pi/2, 0, 0)$ as your initial guess.

- (d) You will write a robot simulator to test the kinematic task-space feedforward plus feedback control law (13.37) tracking the end-effector trajectory

defined by the path

$$X_d(s) = \begin{bmatrix} \sin(s\pi/2) & 0 & \cos(s\pi/2) & s \\ 0 & 1 & 0 & 0 \\ -\cos(s\pi/2) & 0 & \sin(s\pi/2) & 0.491 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad s \in [0, 1],$$

and the time scaling

$$s(t) = \frac{3}{25}t^2 - \frac{2}{125}t^3, \quad t \in [0, 5].$$

In other words, the total time of the motion is 5 s.

Your program should take the trajectory and initial configuration of the robot as input, in addition to the control gains and any other parameters you see fit. In this exercise, due to initial error, the initial configuration of the robot is not on the path: $q_0 = (\phi_0, x_0, y_0) = (-\pi/8, -0.5, 0.5)$ and $\theta_0 = (0, -\pi/4, \pi/4, -\pi/2, 0)$.

Your main program loop should run 100 times per simulated second, i.e., each time step is $\Delta t = 0.01$ s, for a total of 500 time steps for the simulation. Each time through the loop, your program should:

- Calculate the desired configuration X_d and twist $\mathcal{V}_d$ at the current time.
- Calculate the current configuration error $X_{\text{err}} = (\omega_{\text{err}}, v_{\text{err}})$ and save X_{err} in an array for later plotting.
- Evaluate the control law (13.37) to find the commanded wheel speeds and joint velocities $(u, \dot{\theta})$.
- Step the simulation of the robot's motion forward in time by Δt to find the new configuration of the robot. You may use a simple first-order Euler integration for the arm: the new joint angle is just the old angle plus the commanded joint velocity multiplied by Δt . To calculate the new configuration of the chassis, use odometry from Section 13.4, remembering that the change in wheel angles during one simulation step is $u\Delta t$.

You are encouraged to test the feedforward portion of your controller first. A starting configuration on the path is approximately $q_0 = (\phi_0, x_0, y_0) = (0, -0.526, 0)$ and $\theta_0 = (0, -\pi/4, \pi/4, -\pi/2, 0)$. Once you have confirmed that your feedforward controller works as expected, add a nonzero proportional gain to get good performance from the configuration with initial error. Finally, you can add a nonzero integral gain to see transient effects such as overshoot.

After the simulation completes, plot the six components of X_{err} as a function of time. If possible, choose gains K_p and K_i such that it is possible to see typical features of a PI velocity controller: a little bit of overshoot and oscillation and eventually nearly zero error. You should choose control gains such that the 2% settling time is one or two seconds, so the transient response is clearly visible. If you have the visualization tools available, create a movie of the robot's motion corresponding to your plots.

- (e) While retaining stability, choose a set of different control gains that gives a visibly different behavior of the robot. Provide the plots and movie and comment on why the different behavior agrees, or does not agree, with what you know about PI velocity control.

Exercise 13.34 One type of wheeled mobile robot, not considered in this chapter, has three or more conventional wheels which are all individually steerable. Steerable conventional wheels allow the robot chassis to follow arbitrary paths without relying on the passive sideways rolling of mecanum wheels or omniwheels.

In this exercise you will model a mobile robot with four steerable wheels. Assume that each wheel has two actuators, one to steer it and one to drive it. The wheel locations relative to the chassis frame $\{b\}$ mimic the case of the four-wheeled robot in Figure 13.5: they are located at the four points $(\pm\ell, \pm w)$ in $\{b\}$. The steering angle θ_i of wheel i is zero when it rolls in the $+\hat{x}_b$ -direction for a positive driving speed $u_i > 0$, and a positive rotation of the steering angle is defined as counterclockwise on the page. The linear speed at wheel i is ru_i , where r is the radius of the wheel.

- (a) Given a desired chassis twist $\mathcal{V}_b$, derive equations for the four wheel steering angles θ_i and the four wheel driving speeds u_i . (Note that the pair (θ_i, u_i) yields the same linear motion at wheel i as $(-\theta_i, -u_i)$.)
- (b) The “controls” for wheel i are the steering angle θ_i and the driving speed u_i . In practice, however, there are bounds on how quickly the wheel steering angle can be changed. Comment on the implications for the modeling, path planning, and control of a steerable-wheel mobile robot.